

THE THREE SPEEDS OF PHYSICAL AI & THE LATENCY STARVATION BOTTLENECK

How multi-step action chunking bridges the gap between neural inference latency and 1 kHz motor control

SYSTEM 2: SEMANTIC INTENT (0.5 - 2 Hz)

VLM Cognition / Expiring Leases

- Workload: Vision-Language grounding
- Latency budget: 500 – 2000 ms
- Output: Goal affordance lease L_{intent}
- Platform: Linux MPU / Cloud

SYSTEM 1.5: ACTION CHUNKING (20 - 50 Hz)

Diffusion Policy / Trajectory Decoders

- Workload: Multi-step trajectory synthesis
- Latency budget: 20 – 50 ms (P99 $\leq$ 45 ms)
- Output: Action chunks $A_{\{t:t+H\}} (H=16)$
- Platform: Embedded NPU / GPU

SYSTEM 1: SAFETY REFLEX (1000 Hz)

Control Barrier Functions / PWM Driver

- Workload: Mathematical safety enforcement
- Latency budget: < 1.0 ms (Zero tail jitter)
- Output: Gate driver motor PWM currents
- Platform: Deterministic Bare-Metal MCU

✗ SINGLE-STEP POLICY: LATENCY STARVATION & CHATTER

Infer a₀

STARVATION (49ms)

Infer a₁

STARVATION (59ms)

a₀ (1ms)

a₁

- **Physical Defect:** Motor executes 1 step (1 ms), then starves for 49 ms waiting for next token.
- **Zero-Order Hold Jumps:** Discrete position steps command infinite acceleration impulses.
- **Consequence:** Violent mechanical vibration, gearbox fatigue, and motor coil overheating.

✓ SYSTEM 1.5 ACTION CHUNKING: CONTINUOUS EXECUTION RESERVOIR

Infer Action

Chunk Reservoir: a₀, a₁, ..., a₁₅ (16 Waypoints / 320 ms)

- **Temporal Amortization:** 50 ms inference generates 320 ms of smooth physical trajectory.
- **Jitter Immunity:** Motor queue never starves even if inference tail spikes from 40 ms to 120 ms.
- **1 kHz Spline Interpolation:** MCU evaluates smooth C² quintic polynomials every 1 ms tick.